# GOLD STANDARD PROMPT REVISION

OASIS_GOLD_STANDARD_PROMPT · oasis_gold_standard.json Generation · Step 6

*All bugs sourced from SYN_0001 validation report + arithmetic verification*

| Metric | Value |
|---|---|
| **Root cause** | OASIS_GOLD_STANDARD_PROMPT inherits corrupted gap values AND has independent generation errors. Even after correcting the gap prompt, the gold standard prompt has 7 additional bugs that must be fixed. |
| **Upstream dependency** | gap_context feeds gold standard — gap prompt bugs fixed first (previous revision). Gold prompt bugs listed here are independent of gap. |
| **Bugs in current gold prompt** | 7 (2 structural, 1 field definition, 2 arithmetic, 1 propagation, 1 placeholder) |
| **Fields wrong in SYN_0001 gold** | 8 missing admission GG codes + 4 wrong N0415 flags + C0400 arithmetic + PHQ_MOOD_INTERVIEW + A1010 = 15+ corrupted fields |
| **New placeholder** | {medication_json} replaces {medication_summary} for deterministic N0415 |
| **Downstream impact** | Every wrong gold field = one corrupt training example fed to the OAPI optimizer |

# 1. Complete Bug Inventory — Gold Standard Prompt

| # | Bug | Effect on oasis_gold_standard.json | Severity |
|---|---|---|---|
| **BUG-1** | **A1010 field: street address instead of Race/Ethnicity flags** | A1010 in OASIS-E1 is a Race field (multi-select flags). Gold has a street address there. Optimizer reads A1010 expecting a race code, receives a street address — corrupted training slot. | **HIGH — wrong field value** |
| **BUG-2** | **PHQ_MOOD_INTERVIEW key in output** | Not an OASIS-E1 item code. Optimizer does exact key matching — this key either errors out or silently wastes a training slot. | **HIGH — invalid key in gold** |
| **BUG-3** | **{medication_summary} is text, not structured JSON — N0415 flags wrong** | Prompt receives narrative med summary. LLM can't reliably parse drug names to binary flags → N0415E=0 (Warfarin missed), N0415F=0 (Digoxin missed). 4+ wrong N0415 flags. | **CRITICAL — wrong training signal** |
| **BUG-4** | **GG admission codes (X1) missing for 8 activities** | Gap had generic GG0130/GG0170 root codes. Gold prompt mapping rule lists only A–K for GG0170, misses D1/E1/F1 and all of A1–F1 except G1. 8 mandatory admission codes absent from gold. | **CRITICAL — missing training slots** |
| **BUG-5** | **BIMS C0400 grouping total not re-verified in gold** | Gap had C0400=5 while C0400A+B+C=6. Gold prompt propagates gap value verbatim without checking arithmetic → C0400 mismatch persists into gold standard. | **HIGH — arithmetic propagation** |
| **BUG-6** | **PHQ D0160 not re-verified in gold prompt** | Gap had D0160=4 (wrong — should be 2 on negative screen). Gold prompt propagates without checking PHQ-2 gate arithmetic. | **HIGH — arithmetic propagation** |
| **BUG-7** | **Authority rule too weak — gold re-derives values that gap has already answered** | Current rule says 'use Step 4 DIRECTLY as the basis' but no enforcement. LLM re-derives clinical values and diverges (20 mismatches found between gold and gap in SYN_0001). | **HIGH — consistency violations** |

# 2. All Changes — Annotated Comparison Table

| Section / Field | Status | What Changed & Why | OLD | NEW |
|---|---|---|---|---|
| **A1010 in output** | **FIXED** | A1010 is the OASIS-E1 Race field (multi-select flags). The current prompt has no A1010 rule and the LLM fills it with whatever narrative text is nearby (e.g. street address). Added explicit prohibition: A1010 must never contain address data; if included in the batch, it should receive the correct race flag value from the referral. | "A1010": "2847 Maple Grove Lane..." | A1010 = Race flag (e.g. "5" = White). Address must NOT appear in gold standard. |
| **PHQ_MOOD_INTERVIEW** | **REMOVED** | Not an OASIS-E1 item code. The output rules previously permitted free-text keys, and the LLM consistently adds PHQ_MOOD_INTERVIEW to the gold standard output. New OUTPUT SCOPE RULE explicitly lists it as forbidden. | "PHQ_MOOD_INTERVIEW": "completed" | (removed — not an OASIS code) |
| **{medication_summary}** | **FIXED** | Old placeholder was a narrative text summary. LLM cannot reliably extract drug class membership from prose to set binary N0415 flags. Replaced with {medication_json} (structured JSON with is_high_risk boolean per drug). Same approach as corrected gap prompt. | {medication_summary} (narrative text) | {medication_json} (structured JSON with is_high_risk flags) |
| **GG0170 admission codes** | **FIXED** | Old propagation rule listed 'A→A1/A2, B→B1/B2, C→C1/C2, D→D1/D2, E→E1/E2, F→F1/F2, I→I1/I2...' — BUT the gap file had generic GG0170 root codes, so the LLM only emitted the codes it could match. Result: GG0170A1–F1 (6 admission codes) are completely absent from the gold standard. New rule enumerates all 16 X1 codes by name with 'MUST emit' markers. | GG0170 'A'→A1/A2, 'B'→B1/B2... (6 admission codes missing in output) | GG0170A1 Roll L/R ← MUST emit GG0170B1 Sit to Lying ← MUST emit GG0170C1 Lying to Sitting ← MUST emit GG0170D1/E1/F1 ← MUST emit + all X2 discharge goals |
| **GG0130D1 / GG0130E1** | **FIXED** | Upper Body Dressing and Lower Body Dressing admission codes absent from gold standard because gap had generic GG0130 | GG0130D1: MISSING GG0130E1: MISSING | GG0130D1 / GG0130D2 = Upper Body Dressing ← MUST emit GG0130E1 / GG0130E2 = Lower Body Dressing ← MUST emit |

| | | | | |
|---|---|---|---|---|
| | | root codes for these activities. New rule explicitly names D1 and E1 with 'frequently missing — MUST emit' annotation. | | |
| **BIMS arithmetic verify** | **ADDED_RULE** | Gold prompt propagated C0400=5 from gap even though C0400A+B+C=6. New MANDATORY RULE 2 requires the gold prompt to verify BIMS two-level arithmetic and silently correct grouping totals if they disagree with sub-scores. | (absent — no arithmetic verification) | MANDATORY RULE 2: verify C0300=C0300A+B+C, C0400=C0400A+B+C, C0500=C0200+C0300+C0400. Correct and note in rationale if gap value is wrong. |
| **PHQ D0160 verify** | **ADDED_RULE** | Gold prompt propagated D0160=4 from gap even though screen=2 (negative) → correct value is 2. New MANDATORY RULE 3 requires verification of PHQ-2 gate and correction if gap D0160 is inconsistent. | (absent — no PHQ arithmetic verify) | MANDATORY RULE 3: verify screen_score, check PHQ-2 gate, recompute D0160 if needed. Correct silently and note in rationale. |
| **Authority rule** | **TIGHTENED** | Old rule said 'use Step 4 values DIRECTLY as the basis'. LLM interpreted this loosely and re-derived 20 fields independently (mismatches found: B1300, J0520, M1500, M1600, M1610, M1630, M2020, M2030, M2040, etc.). New rule is unambiguous: for every field in gap_context, copy verbatim, rationale must say 'Propagated verbatim from Step 4 gap assessment'. | Use the Step 4 values DIRECTLY as the basis — do NOT re-derive from documents. | For EVERY field in gap context: copy value VERBATIM. Do NOT re-derive, re-interpret, or override. Rationale MUST state 'Propagated verbatim from Step 4 gap assessment.' |
| **N0415 derivation rule** | **ADDED_RULE** | No explicit N0415 rule existed in gold prompt. LLM guessed from narrative and produced N0415E=0 (missed Warfarin as anticoagulant), N0415F=0 (missed Digoxin), N0415A/B/C/D/G missing entirely. New MANDATORY RULE 4 provides full drug class → sub-code mapping table, derived from medication_json. | (absent — no N0415 rule) | MANDATORY RULE 4: read medication json, derive N0415A-N0415I flags. Warfarin→N0415B='1', Digoxin→N0415F='1'. All 9 sub-codes MUST appear in output. |
| **Output scope rule** | **ADDED** | No upfront scope constraint. LLM routinely adds PHQ_MOOD_INTERVIEW and occasionally other | (absent) | □ OUTPUT SCOPE RULE block: PHQ_MOOD_INTERVIEW, ALLERGIES, VITAL_SIGNS, CIRCULATORY_HISTORY, |

| | | | | |
|---|---|---|---|---|
| | | EHR narrative keys. New OUTPUT SCOPE RULE block at top of prompt names forbidden keys and states 'skip it entirely'. | | MENTAL STATUS, etc. are FORBIDDEN. Skip non-OASIS codes entirely. |

# 3. GG Code Completeness — Before vs. After

The gold standard in SYN_0001 is missing 8 mandatory admission codes. The root cause is that the gap file had generic root codes (GG0170, GG0130 without suffixes) which the gold prompt could not map to specific activities. The corrected gold prompt enumerates every required code explicitly.

| Activity | X1 (Admission) | X2 (DC Goal) | SYN_0001 Gold Status | After Fix |
|---|---|---|---|---|
| GG0130A — Eating | **GG0130A1** | **GG0130A2** | ✅Present (05) | ✅Emit from gap |
| GG0130B — Oral Hygiene | **GG0130B1** | **GG0130B2** | ✅Present (04) | ✅Emit from gap |
| GG0130C — Shower/Bathe | **GG0130C1** | **GG0130C2** | ✅Present (02) | ✅Emit from gap |
| GG0130D — Upper Body Dressing | **GG0130D1** | **GG0130D2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0130E — Lower Body Dressing | **GG0130E1** | **GG0130E2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0130F — Toileting Hygiene | **GG0130F1** | **GG0130F2** | ✅Present (03) | ✅Emit from gap |
| GG0170A — Roll Left/Right | **GG0170A1** | **GG0170A2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0170B — Sit to Lying | **GG0170B1** | **GG0170B2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0170C — Lying to Sitting | **GG0170C1** | **GG0170C2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0170D — Sit to Stand | **GG0170D1** | **GG0170D2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0170E — Chair/Bed Transfer | **GG0170E1** | **GG0170E2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0170F — Toilet Transfer | **GG0170F1** | **GG0170F2** | ❌MISSING | ✅MUST emit — explicit in rule |
| GG0170G — Car Transfer | **GG0170G1** | **GG0170G2** | ✅Present (03) | ✅Emit from gap |
| GG0170RR — Wheel 50 ft | **GG0170RR1** | **(no X2)** | ✅Present (3) | ✅Emit from gap |

# 4. N0415 Flags — Before vs. After

In SYN_0001, only 4 N0415 sub-codes appear in the gold standard (E, F, H, I — all set to '0'). Sub-codes A, B, C, D, G are completely absent. Both Warfarin (anticoagulant → N0415B) and Digoxin (cardiovascular → N0415F) are in the medication list as is_high_risk=true, but are coded as '0' because the prompt had no N0415 rule and no structured medication input.

| Code | Drug Class | Triggered by | SYN_0001 | After Fix |
|---|---|---|---|---|
| **N0415A** | Antipsychotic | haloperidol, quetiapine, risperidone, olanzapine | **MISSING** | 0 (none in med list) |
| **N0415B** | Anticoagulant | warfarin ✓ is_high_risk=true in med list | **MISSING** | 1 ← CORRECTED |
| **N0415C** | Antibiotic | any systemic antibiotic | **MISSING** | 0 (none in med list) |
| **N0415D** | Antiplatelet | aspirin ≥325mg, clopidogrel, ticagrelor | **MISSING** | 0 (none in med list) |
| **N0415E** | Hypoglycemic | insulin, metformin, glipizide, etc. | 0 (wrong) | 0 (no insulin — CHF) |
| **N0415F** | Cardiovascular | digoxin ✓ is_high_risk=true in med list | **0 (WRONG)** | 1 ← CORRECTED |
| **N0415G** | Diuretic | furosemide, torsemide, bumetanide, HCTZ | **MISSING** | 1 (furosemide in CHF) |
| **N0415H** | Opioid | oxycodone, morphine, hydromorphone, fentanyl | 0 | 0 (no opioids) |
| **N0415I** | None of above | set 1 only if ALL A-H are 0 | 0 | 0 (B, F, G are 1) |

# 5. Arithmetic Verification Rules — Side by Side

## 5.1 BIMS C0400 Propagation Bug

| ✗ CURRENT (no verification) | ✓ CORRECTED (verifies before writing) |
|---|---|
| **Gold prompt receives gap_context with:**<br>`C0400A = 2, C0400B = 2, C0400C = 2`<br>`C0400 = 5 ← wrong in gap file`<br><br>Rule says: 'use Step 4 values DIRECTLY'<br>Result: C0400=5 copied verbatim into gold<br>But C0400A+B+C = 6 ≠ 5 → arithmetic inconsistency preserved in gold standard | **MANDATORY RULE 2 — verify before writing:**<br>`Step A: C0300 = C0300A+C0300B+C0300C`<br>`Step B: C0400 = C0400A+C0400B+C0400C`<br>`Step C: C0500 = C0200+C0300+C0400`<br><br>If gap C0400 ≠ computed value:<br>→ Use computed value in gold standard<br>→ Rationale: 'C0400 corrected: sub-scores sum to 6, gap had 5' |

## 5.2 PHQ D0160 Propagation Bug

| ✗ CURRENT (no verification) | ✓ CORRECTED (re-checks gate) |
|---|---|
| **Gap file has:**<br>`D0150A1=1, D0150B1=1 → screen=2 (negative)`<br>`D0160=4 ← wrong (should be 2)`<br><br>RULE 6 in current prompt says:<br>'D0160 = only A2+B2 (if present)'<br>But no instruction to verify gap_context D0160<br>Result: D0160=4 propagated into gold standard | **MANDATORY RULE 3 — re-verify:**<br>`screen = D0150A1 + D0150B1`<br>`if screen < 3 (negative):`<br>`  D0160 = (A2 if A1=1 else 0)`<br>`  + (B2 if B1=1 else 0)`<br><br>If gap D0160 ≠ computed value:<br>→ Use correct computed value in gold<br>→ Rationale: 'D0160 corrected: negative screen (score 2), correct total = 2' |

# 6. Authority Rule — Gap vs. Re-Derivation

20 mismatches were found between gold standard and gap answers in SYN_0001. Some are legitimate (gold is more clinically accurate), but most represent the LLM ignoring the gap answers and re-deriving from documents. The new authority rule eliminates re-derivation for any field present in gap_context.

| Field | Gold | Gap | Status | Verdict |
|---|---|---|---|---|
| **D0160** | 2 | 4 | **Gold CORRECT** | Gap wrong (PHQ-2 gate violation). Gold prompt must verify and correct — new RULE 3 handles this. |
| **B1300** | 0 | 2 | **Investigate** | Gap=2 (moderate health literacy issue) vs Gold=0. Clinical docs support moderate literacy needs — gap may be correct. Authority rule: gold must propagate gap value if gap is corrected upstream. |
| **J0520** | 0 | 1 | **Gold CORRECT** | Pain not affecting sleep documented in scribe. Gap answer appears wrong. Gold re-derivation was correct here — shows authority rule must apply only when gap is accurate. |
| **M1005** | 03062026 | 2026-03-06 | **Format only** | Same date, different format. Gold uses MMDDYYYY (correct). Gap uses ISO. Gold formatting rule is correct — no content mismatch. |
| **M1500** | 2 | 1 | **Gold CORRECT** | CHF patient with dyspnea with ADLs → M1500=2. Gap=1 (dyspnea with exertion only) appears under-coded. Gold correctly re-derived. |
| **M1600** | 1 | 2 | **Investigate** | M1600=1 (treated for UTI) vs M1600=2. Neither is clearly documented for CHF patient. Both may be wrong — clinical review needed. |
| **M2020** | 1 | 2 | **Gold likely correct** | M2020=1 (able to take correctly) vs gap=2 (unable). Scribe documents med compliance. Gap over-coded. |
| **M1740** | 7 | ['None'] | **Gold CORRECT** | Gap has wrong type (array). Gold='7' (none of above) is correct coded value. |
| **M1028** | 0 | {dict} | **Gold CORRECT** | Gap has wrong type (dict of diagnoses). M1028 is 0=No/1=Yes for PVD. Gold=0 correct format. |

Key insight: the authority rule (copy gap verbatim) only works correctly AFTER the gap prompt fixes are applied. For arithmetic errors (D0160, C0400), the gold prompt must still verify and correct — this is why MANDATORY RULES 2 and 3 were added to the gold prompt independently.

# 7. Implementation — Placeholder Change

{medication_summary} is replaced by {medication_json}. The pipeline call site must be updated.

## Pipeline Call Site Update

```
# OLD call site
prompt = OASIS_GOLD_STANDARD_PROMPT.format(
    archetype=archetype,
    diagnosis_context=diagnosis_context,
    has_scribe=has_scribe,
    section_name=section_name,
    referral_text=referral_text,
    scribe_section=scribe_section,
    medication_summary=medication_summary_text,  # ← narrative text
    gap_context=gap_context,
    field_codes_json=field_codes_json,
)

# NEW call site
import json
med_json_str = json.dumps(medication_list_dict, indent=2)

prompt = OASIS_GOLD_STANDARD_PROMPT.format(
    archetype=archetype,
    diagnosis_context=diagnosis_context,
    has_scribe=has_scribe,
    section_name=section_name,
    referral_text=referral_text,
    scribe_section=scribe_section,
    medication_json=med_json_str,               # ← structured JSON
    gap_context=gap_context,
    field_codes_json=field_codes_json,
)
```

## Recommended Post-Processing Validator

Even with corrected prompts, a Python validator on the gold standard output catches any remaining arithmetic slip.

```
def validate_gold_standard(gold: dict) -> list[str]:
    errors = []
    # Forbidden keys
    forbidden = {'PHQ_MOOD_INTERVIEW', 'ALLERGIES', 'VITAL_SIGNS', 'CIRCULATORY_HISTORY'}
    for k in gold:
        if k in forbidden:
            errors.append(f'Forbidden key in gold: {k}')
    # BIMS arithmetic
    if all(k in gold for k in ['C0300A','C0300B','C0300C','C0300']):
        exp = int(gold['C0300A'])+int(gold['C0300B'])+int(gold['C0300C'])
        if int(gold['C0300']) != exp: errors.append(f'C0300 mismatch: {gold["C0300"]} !=
{exp}')
    if all(k in gold for k in ['C0400A','C0400B','C0400C','C0400']):
        exp = int(gold['C0400A'])+int(gold['C0400B'])+int(gold['C0400C'])
        if int(gold['C0400']) != exp: errors.append(f'C0400 mismatch: {gold["C0400"]} !=
{exp}')
```

```
    if all(k in gold for k in ['C0200','C0300','C0400','C0500']):
        exp = int(gold['C0200'])+int(gold['C0300'])+int(gold['C0400'])
        if int(gold['C0500']) != exp: errors.append(f'C0500 mismatch: {gold["C0500"]} !=
{exp}')
    # PHQ gate
    a1 = int(gold.get('D0150A1', 0)); b1 = int(gold.get('D0150B1', 0))
    if a1+b1 < 3:
        a2 = int(gold.get('D0150A2',0)) if a1==1 else 0
        b2 = int(gold.get('D0150B2',0)) if b1==1 else 0
        if str(gold.get('D0160')) != str(a2+b2):
            errors.append(f'D0160 gate violation: got {gold["D0160"]}, should be {a2+b2}')
    # N0415 completeness
    for sub in 'ABCDEFGHI':
        if f'N0415{sub}' not in gold:
            errors.append(f'N0415{sub} missing from gold standard')
    # GG admission completeness
    must_have =
['GG0130D1','GG0130E1','GG0170A1','GG0170B1','GG0170C1','GG0170D1','GG0170E1','GG0170F1']
    for k in must_have:
        if k not in gold: errors.append(f'{k} missing from gold standard')
    return errors
```

## What's Now Fixed vs. Still Pending

| ✅ Fixed by this revision (gold prompt) | ⬜ Not covered — separate work items |
|---|---|
| • A1010 field definition error removed<br>• PHQ_MOOD_INTERVIEW key banned from output<br>• N0415 A–I all derived from medication JSON<br>• N0415B=1 (Warfarin) and N0415F=1 (Digoxin) corrected<br>• GG0130D1/E1 explicitly required in rule<br>• GG0170A1–F1 all explicitly required in rule<br>• BIMS C0400 arithmetic verified and corrected<br>• PHQ D0160 re-verified against PHQ-2 gate<br>• Authority rule hardened — gap propagated verbatim<br>• Output scope rule added — non-OASIS keys banned | • metadata.json validation_status update logic (pipeline code)<br>• Physician NPI fictional-range enforcement (referral generation, not gold prompt)<br>• M0080 NPI field in gold updated to fictional range in pipeline code<br>• H-6 M1330 wound consistency (CHF archetype — gap needs clinical review)<br>• GG_CONSISTENCY_REPAIR_PROMPT — can now be simplified (gold no longer needs repair if gap prompt is correct) |